3. Results

M3 Research Project · Detection-Constrained Power Allocation for Cognitive ISAC Relays under Imperfect CSI (NCC 2027 submission)

Source: submitted NCC 2027 paper, Sec. V-B to V-F, Figs. 2-4. `reported_values_check.csv` recomputes every number below from the files in `data/`.

Figures (`figures/`)

Figure	File	Data	Script
Fig. 2: Protected SAR vs CSI-error ratio ε (60 paired channels)	`Fig2_protected_SAR_vs_CSI_error.pdf`	`main_comparison.csv`, `tuned_baselines.csv`	`plot_protected_sar_vs_eps.py`
Fig. 3: Average SAR vs empirical channel protection, ε = 0.01 (40 channels)	`Fig3_rate_vs_protection.pdf`	`tradeoff.csv`, `main_comparison.csv`	`plot_protection_rate_panels.py`
Fig. 4: Worst-target P_D vs total admitted ST power, ε = 0.01	`Fig4_power_vs_detection.jpg`	`main_comparison.csv`, `tuned_baselines.csv`	`plot_power_vs_pd.py`

To redraw a figure, run its script from `data/` (needs `numpy`, `scipy` and `matplotlib`). For example:

```
cd data
python plot_protected_sar_vs_eps.py
```

Main results

Sensitivity to CSI errors (Fig. 2, Sec. V-B)

- **Proposed design.**
  - With perfect CSI, it reaches 10.67 bit/s/Hz, and it loses rate gradually as ε grows.
  - At ε = 0.01, it retains **8.37** bit/s/Hz, against 4.76 for Hu robust: a **76 % gain**. The tuned Hu robust design gives 5.54 and the adapted worst-case design gives 5.86.
  - At ε = 0.03, it achieves **6.57** bit/s/Hz, against 4.43 (adapted worst-case) and 3.04 (Hu robust).
- **Hu non-robust.**
  - For $\varepsilon <= 1.5 \cdot 10^{-3}$, the tuned floor comes within 3 % of the proposed design.
  - The tuned design reaches zero protected SAR at ε = 0.03.

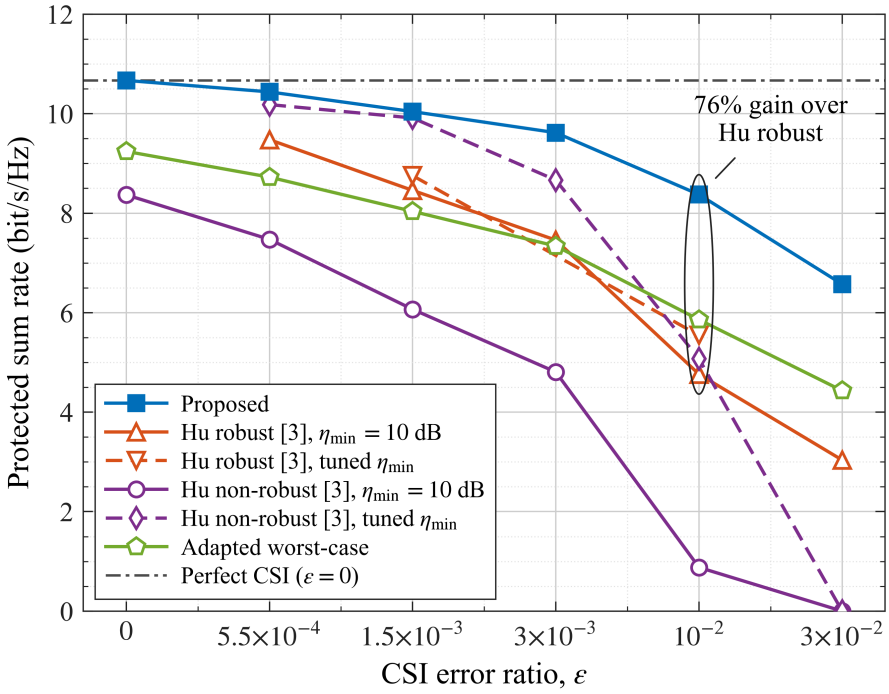


Fig. 2. Protected SAR versus CSI-error ratio ε (mean over 60 paired channels). The dash-dot line is the proposed design with perfect CSI.

Rate at a common protection requirement (Fig. 3, Sec. V-C)

- **Proposed design.**
  - For β = 0.5-0.8, it gives a high average SAR, but it protects only a small fraction of channels.
  - At β = 0.9, it reaches 100 % observed protection with about **8.4** bit/s/Hz.
- **Baselines at 100 % protection:**
  - adapted worst-case: about 7.3 bit/s/Hz (4 dB);
  - Hu robust: about 5.8 bit/s/Hz (4 dB);
  - Hu non-robust: about 5.3 bit/s/Hz (19 dB).
- The proposed design is **15 % above the best benchmark**.

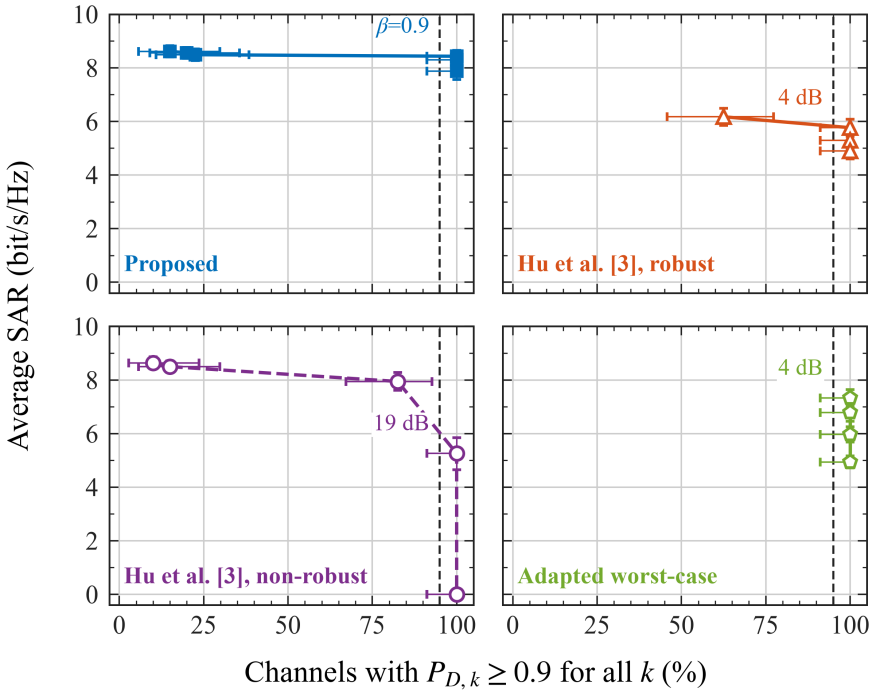


Fig. 3. Average SAR versus empirical channel protection, each scheme swept over its design parameter (ε = 0.01, 40 channels; error bars: 95 % confidence intervals; dashed line: 95 % protection).

Secondary power and detection performance (Fig. 4, Sec. V-D)

- **Proposed design.** Its points lie above P_D = 0.9, many of them close to 0.9. Admitted powers range from about 60 to 390 mW.
- **Untuned non-robust design.** It sits around 300-400 mW, with many P_D values well below 0.9.
- **Robust and adapted worst-case designs.** They keep P_D close to 1, a larger margin that costs rate.

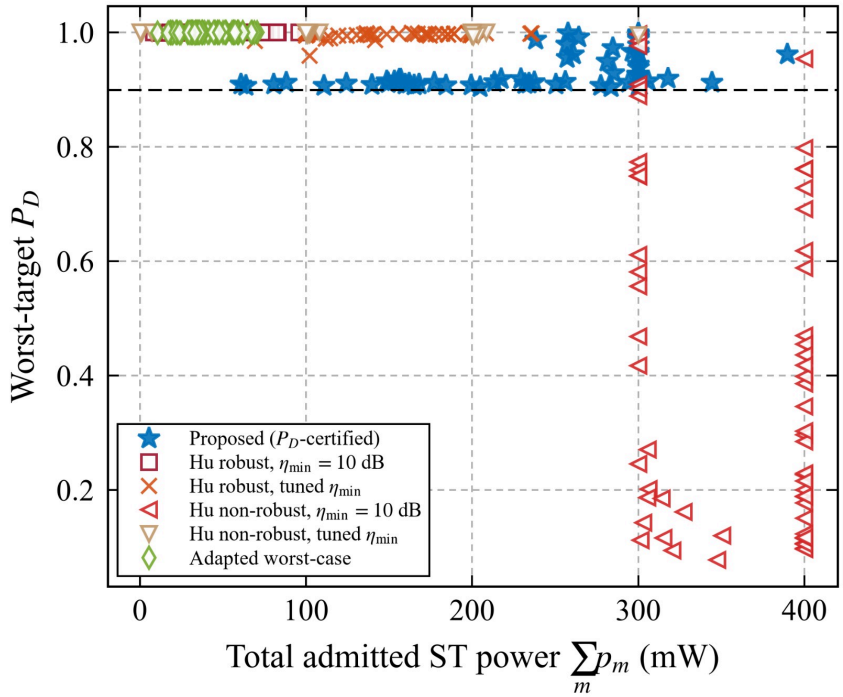


Fig. 4. Worst-target detection probability versus total admitted ST power (ε = 0.01). The dashed line marks P_D = 0.9.

Component comparison (Sec. V-E, `component_study_eps0.01.csv`)

The allocation procedure is fixed and only the radar-protection constraint changes. Setting: 40 paired channels, ε = 0.01.

Protection constraint	`param` in data	Protected SAR (bit/s/Hz)	Channels protected
Nominal SINR floor	F1-S	2.83	≈ 38 %
Gaussian detection model	F1-PG	5.79	≈ 68 %
Worst-case SINR rule	F1-B	6.04	100 %
Proposed (P_D, compound-Gaussian)	F1-P	**8.43**	100 %

The proposed design is +2.39 bit/s/Hz, or **+39.6 %**, above the worst-case SINR rule.

Clutter sensitivity (Sec. V-E, `clutter_shape_endpoints.csv`)

- The protected rate goes from 8.44 bit/s/Hz at ν = 50 to 8.39 at ν = 0.3, about -0.6 %.
- All evaluated channels stay protected.
- This holds over the two tested shapes. It is not a claim of invariance to any clutter distribution.

Scope of the protection requirement (Sec. V-F, `main_comparison.csv`, ε = 0.01)

- **Per error draw.** The proposed design meets P_D >= 0.9 on 71 % of the evaluated error draws. The worst-case designs meet it on 100 %.
- **Worst-case SAR over the bounded error sets.** The proposed design gives 2.30 bit/s/Hz and Hu robust gives 2.86.
- A higher error-averaged rate therefore does not imply better worst-case performance.

Data files (`data/`)

File	Contents	Used for
`main_comparison.csv`	4 schemes at their default settings × 6 ε values × 60 channels	Fig. 2, Fig. 3 (default points), Fig. 4, Sec. V-F
`tuned_baselines.csv`	Hu robust and Hu non-robust at their tuned floors × 60 channels	Fig. 2, Fig. 4
`tradeoff.csv`	Each scheme swept over its design parameter, ε = 0.01, 40 channels	Fig. 3
`component_study_eps0.01.csv`	The four protection-constraint variants, ε = 0.01, 40 channels	Sec. V-E
`clutter_shape_endpoints.csv`	Proposed design at ν = 50 and ν = 0.3, ε = 0.01, 40 channels	Sec. V-E